



IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are **entered** in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)

2	8	3	9	7	6	8
N	N	N	N	N	N	N

Date of Birth (DD/MM/YYYY)

0	9	1	0	3	1	2	0	0	4
D	D		M	M		Y	Y	Y	Y

COMPULSORY Q1.

Do not mark any other boxes.

☒ Q1	☐ DO NOT USE	☐ DO NOT USE	☐ DO NOT USE	☐ DO NOT USE	☐ DO NOT USE	☐ DO NOT USE
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You may use both sides but only answer **ONE** question per sheet

Q1 (a)

For Glasgow: $L_G = 55^\circ 52' 19'' = 55.871944^\circ$
 $\lambda_G = -4^\circ 25' 59'' = -4.433056^\circ$

For Melbourne: $L_M = -37^\circ 40' 24'' = -37.673333^\circ$
 $\lambda_M = 144^\circ 50' 36'' = 144.843333^\circ$

The radius of curvature at each point:

$$R_E(L_G) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_G}} = 6,392,816.367 \text{ m}$$

$$R_E(L_M) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_M}} = 6,386,126.095 \text{ m}$$

Therefore, Glasgow is at:

$$x_{G\beta}^e = R_E(L_G) \cos L_G \cos \lambda_G = 3,575,923.739 \text{ m}$$

$$y_{G\beta}^e = R_E(L_G) \cos L_G \sin \lambda_G = -277,227.722 \text{ m}$$

$$z_{G\beta}^e = (1 - e^2) R_E(L_G) \sin L_G = 5,256,456.151 \text{ m}$$

Melbourne is at:

$$x_{M\beta}^e = R_E(L_M) \cos L_M \cos \lambda_M = -4,132,600.514 \text{ m}$$

$$y_{M\beta}^e = R_E(L_M) \cos L_M \sin \lambda_M = 2,910,550.658 \text{ m}$$

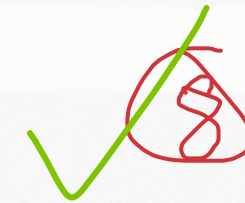
$$z_{M\beta}^e = (1 - e^2) R_E(L_M) \sin L_M = -3,876,808.892 \text{ m}$$

From $\langle \hat{a}, \hat{b} \rangle = \frac{a \cdot b}{|a||b|} \cos \theta$

$$\Rightarrow \theta = \arccos\left(\frac{a \cdot b}{|a||b|}\right) = 152.519582^\circ$$

Therefore:

$$\begin{aligned} \text{distance} &= \text{Earth Circumference} \times \frac{\theta}{360^\circ} \\ &= 40030.2 \text{ km} \times \frac{152.519582^\circ}{360^\circ} \\ &= 16859.4 \text{ km} \end{aligned}$$



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Q1 Continued

You may use both sides but only answer **ONE** question per sheet

(b). The navigation equation resolved in ECEF is:

$$\underline{a}_{ib}^e = \underline{f}_{ib}^e + \underline{y}_b^e = C_b^e \underline{f}_{ib}^b + \underline{y}_b^e$$

To get $\underline{\dot{v}}_{eb}^e$,

$$\underline{r}_{eb}^e = \underline{r}_{ib}^e - \underline{r}_{ie}^e = \underline{r}_{ib}^e$$

$$\Rightarrow \underline{\ddot{r}}_{eb}^e = \underline{\ddot{r}}_{ib}^e$$

$$= -(\Omega_{ie}^e \Omega_{ie}^e + \dot{\Omega}_{ie}^e) \underline{r}_{ib}^e - 2\Omega_{ie}^e \underline{\dot{r}}_{ib}^e + \underline{a}_{ib}^e$$

Since $\underline{v}_{eb}^e = \underline{\dot{r}}_{eb}^e \Rightarrow \underline{\dot{v}}_{eb}^e = \underline{\ddot{r}}_{ib}^e$, and $\underline{\Omega}_{ie}^e = 0$

$$= -[\Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e - 2\Omega_{ie}^e \underline{\dot{r}}_{ib}^e + \underline{a}_{ib}^e]$$

By substituting $\underline{a}_{ib}^e$:

$$\underline{\dot{v}}_{eb}^e = C_b^e \underline{f}_{ib}^b + \underline{y}_b^e - \Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e - 2\Omega_{ie}^e \underline{\dot{r}}_{ib}^e$$

$$= C_b^e \underline{f}_{ib}^b + \underline{g}_b^e(\underline{r}_b^e) - 2\Omega_{ie}^e \underline{\dot{r}}_{ib}^e$$

where $\underline{g}_b^e(\underline{r}_b^e) = \underline{y}_b^e - \Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e$.

Also, notice that $\underline{r}_{ib}^e = \underline{r}_{ie}^e + \underline{r}_{eb}^e = \underline{r}_{eb}^e \Rightarrow \underline{\dot{r}}_{ib}^e = \underline{\dot{r}}_{eb}^e = \underline{\dot{v}}_{eb}^e$
we have:

$$\underline{\dot{v}}_{eb}^e = C_b^e \underline{f}_{ib}^b + \underline{g}_b^e(\underline{r}_b^e) - 2\Omega_{ie}^e \underline{v}_{eb}^e$$

(c).

From $\underline{\dot{v}}_{eb}^e = C_b^e \underline{f}_{ib}^b + \underline{g}_b^e(\underline{r}_b^e) - 2\Omega_{ie}^e \underline{v}_{eb}^e$:

$$\Rightarrow C_b^e \underline{f}_{ib}^b = \underline{\dot{v}}_{eb}^e - \underline{g}_b^e(\underline{r}_b^e) + 2\Omega_{ie}^e \underline{v}_{eb}^e$$

$$\Rightarrow \underline{f}_{ib}^b = C_e^b \underline{\dot{v}}_{eb}^e - C_e^b \underline{g}_b^e(\underline{r}_b^e) + 2C_e^b \Omega_{ie}^e \underline{v}_{eb}^e$$

$$= \underline{a}_{eb}^b - \underline{g}_b^b(\underline{r}_b^e) + 2C_e^b \Omega_{ie}^e \underline{v}_{eb}^e$$

Here: $\underline{a}_{eb}^b = \underline{a}_{ib}^b - \underline{a}_{ie}^b$

and: $\underline{a}_{ib}^b = \underline{\dot{v}}_{ib}^b + \Omega_{ib}^b \underline{v}_{ib}^b$

$\underline{a}_{ie}^b = 0$.

~~and $\underline{a}_{ie}^b = 0$~~

Then: $\underline{f}_{ib}^b = \underline{\dot{v}}_{ib}^b + \Omega_{ib}^b \underline{v}_{ib}^b - \underline{g}_b^b(\underline{r}_b^e) + C_e^b \Omega_{ie}^e \underline{v}_{eb}^e$

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